Self-Written Evidence: Error Propagation When Accepted LLM Answers Are Written Back to a Retrieval Corpus

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Abstract

Retrieval-augmented support systems store the answers their users accept, so that later questions can be answered from them. A wrongly accepted answer then becomes evidence, and answers generated from it can be accepted and stored in turn. We measure this human-gated loop on real repeat questions from four CQADupStack forums, with community duplicate links as ground truth, three retrievers, 48,026 generated answers and an independent judge that checks each answer against the human solution of its source thread, while a simulated user accepts wrong answers at a swept rate. Write-back can fix only 3% to 20% of questions, and it fixes a few percent of those while breaking a similar share of correct answers when users accept most wrong answers; at that point no configuration differs significantly from static retrieval. With the source thread and the length held fixed, stored generated answers displace the correct thread more often than the human answers they paraphrase, mainly when the answer is stored without its question, and the effect on outcomes is comparable in size to the effect of write-back itself. Along chains of answers generated from answers, the share that still states the original solution falls in every configuration, from 76% to 58% pooled over served answers, but the change in correct answers attributable to deep entries is indistinguishable from zero, also in a longer stream with paraphrased repeats in which deep entries serve one answer in six by the end. In a factorial over the prompt, dropping the instruction to answer from the evidence widens this fall by +14.1 points in 10 of 12 configurations, an effect consistent in direction but not significant after correction for multiple tests, while the length limit does not affect it. Neither a first-order nor a depth-aware estimate predicts the crossover acceptance rate for the main entry design. Reopening wrong answers, even one round late or with false reopens, keeps the gain and improves the outcome at high acceptance in some configurations; capping depth, regenerating from the source and down-weighting stored entries do not do better.

1 Introduction

Support systems built on retrieval-augmented generation (RAG) (Lewis et al., 2020) increasingly learn from their own use. When a user accepts an answer, the question and the generated answer are written back into the knowledge base, and the next similar question can retrieve that entry and generate from it (Liu et al., 2026; Wang et al., 2026a; Consortium for Service Innovation, 2016). The loop is attractive because repeat questions are common, and it is risky because users accept wrong answers. A wrongly accepted answer becomes evidence, and an answer generated from it can be accepted and stored in turn, so an error can travel along a chain of answers that were each generated from the previous one.

Two strands of work suggest the risk is real. Neural retrievers prefer model-written text (Dai et al., 2024b), and generated text written back into a corpus displaces human documents over repeated rounds (Chen et al., 2024a; Yu et al., 2026; Druck & Smith, 2026). Agents that store their own outputs propagate their errors (Xiong et al., 2026; Wang et al., 2026b; Chen et al., 2026). These studies either add generated text without any gate, or gate it with an oracle that admits only correct outputs, and few have ground truth for whether a later question was answered correctly. A deployed support system sits in between: a human gate that lets wrong answers through at some rate, and real repeat questions whose correct answers are known.

Guthikonda (2026) measured this human-gated loop when the stored entry is only a pointer to an existing thread, and found that write-back helps while users reject most wrong answers and hurts beyond a crossover acceptance rate. With pointer entries no text is generated, so errors cannot change an answer's content or travel through chains. Here we store what deployed systems store, the generated answer, and ask four questions. Do retrievers prefer stored model-written answers over human text from the same source (H1)? Do errors propagate along chains of answers generated from answers (H2)? Can the acceptance rate at which write-back turns harmful be predicted when entries have depth (H3)? And which countermeasures remove harm without removing the benefit (H4)?

We answer them on four CQADupStack forums with three retrievers, ten random orders and real repeat questions, generating 48,026 answers and checking each against the human solution with an independent judge. Four design choices make the measurement harder to explain away. A control replaces each generated answer by the human answer of the same source thread, cut to the same length, so source bias is tested with the source and length held fixed. A content-aware user accepts according to what the answer says, not only where it came from, so text that loses its solution counts as wrong. Effects are measured against the questions write-back could possibly fix, with uncertainty from a bootstrap over both orders and duplicate families. And countermeasures are tested in realistic form, with delayed and mistaken reopening and rules that act on provenance depth.

Our findings are:

- **The stakes are small and two-sided.** Only 3% to 20% of questions can be fixed by write-back, and it fixes a few percent of them while breaking a similar share of correct answers when users accept most wrong answers (Section 4.1).
- **Retrievers prefer generated answers stored on their own, a little.** The preference shrinks for dense retrievers when the question is stored with the answer, and its effect on outcomes is comparable in size to the effect of write-back itself (Section 4.2).
- **Text degrades along chains in every configuration; measurable harm does not follow within the stream.** In a factorial over the prompt, dropping the instruction to answer from the evidence makes the degradation steeper in most configurations, though not significantly after correction, while the length limit does not matter (Sections 4.3 and 4.6).
- **The crossover is not predictable out of sample for the main design**, by the first-order estimate of Guthikonda (2026) or by a depth-aware one (Section 4.4).
- **Reopening wrong answers keeps the gain even when late or noisy; rules that act on provenance depth do not do better** (Section 4.5).

All code, every generated answer and every judge label are released.

2 Related work

Generated text in retrieval corpora. Neural retrievers rank text written by language models above human text with the same content (Dai et al., 2024b), and paired benchmarks of human passages and model rewrites now measure this source bias (Dai et al., 2024a). Later work traces the bias to low perplexity (Wang et al., 2025), to relevance supervision (Xion & Nejdl, 2026; Huang et al., 2026) and to term statistics in lexical rankers (Huang et al., 2025), and readers show a matching preference for generated contexts (Tan et al., 2024). Once generated text accumulates it displaces human evidence. Chen et al. (2024a) index the answers of a RAG system back into an open-domain corpus for ten rounds and find human passages pushed out of the top ranks, Yu et al. (2026) find that retrieval exposure turns synthetic while answer accuracy looks stable, and Druck & Smith (2026) show that a model retrieving references it wrote itself collapses in most simulations because it over-cites its own text, a bias that survives quality controls but that their observational design cannot separate from content. Write-back of generated answers has also been proposed as an improvement behind automatic validation (Chinthala, 2025) or a reference-based quality score in

community QA (Chen et al., 2025). Our loop differs in that generated text enters only after a simulated user accepts it at a false-acceptance rate we sweep, every stored entry records its provenance depth and, when retrieved, is the only evidence for the next answer, and the human-answer control with the same question, root thread, length and position is applied inside the loop rather than to a static corpus.

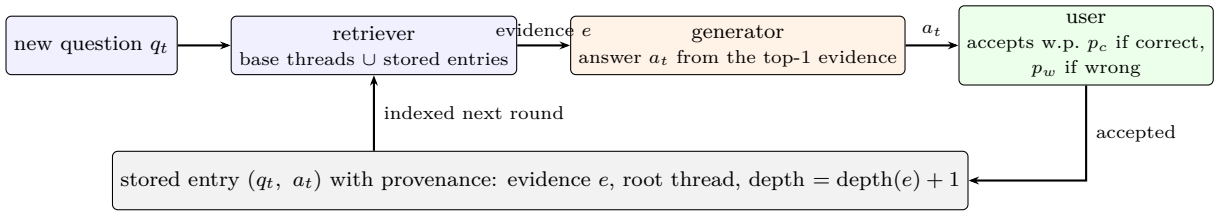
Feedback loops and model collapse. Training generative models on their own outputs erodes the tails of the distribution (Shumailov et al., 2024) and degrades quality or diversity unless fresh real data enters each generation (Alemohammad et al., 2024), whereas accumulating synthetic data beside real data rather than replacing it avoids collapse (Gerstgrasser et al., 2024). When people curate the synthetic samples before retraining, the loop provably optimizes their preference (Ferbach et al., 2024), so a human filter selects for what is accepted rather than for what is correct. Model outputs that re-enter training data amplify dataset biases (Taori & Hashimoto, 2023), predictions that shift the distribution they predict are performative (Perdomo et al., 2020), and recommender loops homogenize behaviour (Chaney et al., 2018), degenerate user interests (Jiang et al., 2019) and escalate a preference for generated content through user clicks (Zhou et al., 2025). A stochastic model of how generated answers are produced and indexed warns that adoption can outpace human verification (Garetto et al., 2025), and activity on Stack Overflow has fallen since ChatGPT was released (del Rio-Chanona et al., 2024), so forum knowledge bases may increasingly hold generated answers. These loops act through retraining, preferences or an open pool of documents. In ours the weights stay fixed, accepted entries accumulate beside the original threads, and acceptance is the only filter, so community duplicate labels let us measure how a wrongly accepted answer is reused as evidence and how far its descendants reach along a provenance chain.

Agent memory and error propagation. Within a single response an early mistake leads a model into further mistakes (Zhang et al., 2024), and agents that store their own outputs carry such errors across tasks. Xiong et al. (2026) show that agents copy the records they retrieve, that stored errors propagate, and that selective addition through a simulated human oracle, which admits only correct executions, improves performance. Errors in stored rules and summaries reproduce themselves through retrieval (Wang et al., 2026b), write-back lets one source fill several retrieval slots through records that point to their parents (Li et al., 2026), and an agent writing to an append-only store either cleans its errors or is captured by them, with majority admission arming a false majority and provenance marks helping only when they track truth (Chen et al., 2026). Proposed remedies include admission by self-consistency (Zhang & Li, 2026), screening and provenance ranking, which Karunanidhi (2026) find of limited use, and dependency-aware cleanup (Choudhary, 2026), while attacks show that injected records steer later outputs (Zou et al., 2025; Chen et al., 2024b; Dong et al., 2025). These studies use agent benchmarks or adversaries, and their gates are either oracles or automatic. We study a support knowledge base whose benign but imperfect user lets wrong answers through at a controlled rate, record the depth of every entry, and compare countermeasures that act on depth, namely a depth cap, regeneration from the root thread and depth-scaled down-weighting, with reopening and uniform down-weighting.

The pointer-only loop. Guthikonda (2026) measured the same human-gated loop on all twelve CQADupStack forums with entries that point to an existing thread and generate no text. Write-back helped while users rejected most wrong answers and hurt beyond a crossover acceptance rate, the harm came from question-to-question similarity, and a first-order estimate from per-entry effects predicted the crossover on held-out orders. We keep that protocol, ground truth and estimate, so the two studies are directly comparable, and change the one thing deployed systems do differently: the stored entry is a generated answer that becomes the evidence for the next one.

3 Setup

Data and stream. We follow the protocol of Guthikonda (2026) so that every number is comparable with the pointer-only loop studied there. We use four CQADupStack forums (Hoogeveen et al., 2015) in their BEIR form (Thakur et al., 2021): android, unix, webmasters and mathematica. The knowledge base is the forum's set of threads. The incoming questions are the held-out questions whose community duplicate links point into the knowledge base, between 1,358 and 1,696 per forum, and questions that share a duplicate



An entry generated from a thread has depth 1. An entry generated from another entry has depth 2 or more, and its evidence chain ends at a root thread. Root correctness asks whether the root thread is a community-marked duplicate of q_t; content correctness additionally asks an independent judge whether a_t states the root thread's solution.

Figure 1: The loop we simulate. Only the user's accept decision is simulated; retrieval, generation and judging are real.

target form a family. Families are what makes write-back useful: a later question can only be helped by an entry written for an earlier member of its own family. Between 53% and 73% of questions belong to a family with more than one question, and the largest family holds up to 15% of a forum's questions, which is why we also report results without it. Each base thread is completed with its accepted answer, or failing that its top-voted answer, fetched by post id from the public Stack Exchange API. Questions arrive in ten random orders. Each order is split into ten rounds, and an entry written in one round becomes retrievable in the next, as in a knowledge base that is re-indexed in batches.

Retrieval and generation. Three retrievers score base threads and stored entries together: bge-small-en-v1.5 (Xiao et al., 2024), all-MiniLM-L6-v2 (Wang et al., 2020; Reimers & Gurevych, 2019) and BM25 (Robertson & Zaragoza, 2009) with the base corpus statistics. The top-1 item is the evidence. If it is a thread, the generator sees the thread and its accepted answer. If it is a stored entry, the generator sees the stored question and the stored answer and never the root thread, as in a deployed system that retrieves its own past answers. The generator is `gpt-5.4-mini` with minimal reasoning and a fixed seed, instructed to answer from the evidence only in at most 120 words. Prompts rejected by the provider's content filter, 1 in total, were answered by an open-weight model and are marked in the released cache. Every answer is generated once per (question, evidence) pair and cached, so every condition that reaches the same pair sees the same text. The loop of generate, judge, index and simulate was iterated until no served pair was missing, which took 48,026 generated answers.

What is stored. An accepted answer is written back as an entry that carries its provenance: the evidence it was generated from, the root thread at the end of that evidence chain, and its depth (Figure 1). We compare three entry designs. The *generated answer* alone; the *question with the generated answer*, the realistic self-written question and answer pair and our main design; and the *generated answer with the root thread's text*, which stores the answer together with its cited source. To test source bias with the source and length held fixed, we add two controls built from the same keys. For every entry, the control replaces the generated answer by the root thread's human answer, cut to the same number of words, with or without the question. Each generated design and its control therefore share the question, the root thread, the length and the position in the stream. They still differ in more than authorship: the generated answer was written in response to the stored question and tends to restate it, while the human extract was not, so the control bounds source bias rather than isolating it. As further reference points we show the two pointer designs of Guthikonda (2026), which store the question alone or the question with the thread text and generate nothing.

Correctness and the user. An answer is *root-correct* when the root thread of its evidence chain is a community-marked duplicate of the question. This is the label used by Guthikonda (2026). It ignores the text of the answer, so on its own it cannot show an error that the generated text introduces. We therefore also label every generated answer with an independent judge, `DeepSeek-V4.1-Flash`, which reads the question, the root thread's human answer and the generated answer and replies YES, PARTLY, NO or ABSTAIN.

An answer is *content-correct* when it is root-correct and the judge says YES. When the root thread has no human answer on record, which is the case for 12% of served answers, content-correctness falls back to root-correctness. The judge labelled every generated answer whose root thread has a human answer, 42,497 in all. On a separate audit of 900 answers it agrees with `gpt-5.4-mini` on 81% and with `gemma-4-26b` on 75% of labels, while those two agree with each other on 78% (Appendix C; Cohen's κ between 0.48 and 0.64). The simulated user accepts a correct answer with probability $p_c = 0.9$ and a wrong one with probability p_w, the swept parameter. The *content-aware* user, our main user model, applies this to content-correctness, so an answer from the right thread that lost its solution is treated as a wrong answer. The *root-label* user applies it to root-correctness, as in Guthikonda (2026).

Countermeasures. We test rules a deployed system could apply. *Reopen* removes a wrongly accepted answer's entry with probability 0.5. In the oracle form used by Guthikonda (2026) the entry is removed before it is ever retrievable, which is equivalent to filtering half of the wrong entries at write time. We therefore add realistic forms in which the entry is retrievable for one or two rounds before it is reopened, and forms in which 10% or 20% of correct entries are reopened by mistake. *Depth cap* stores only answers generated from a base thread, so no entry of depth 2 or more exists. *Regenerate from root* answers from the root thread of a retrieved entry rather than from the entry's text, so the text is always one step from a human answer. *Down-weighting* subtracts λ times the interquartile range of static top-1 scores from every entry's score, either uniformly or multiplied by the entry's depth, with $\lambda \in \{0.5, 1\}$.

Measures and statistics. Δ is the change in the share of questions answered correctly over the stream against static retrieval, in percentage points, using content-correctness for the content-aware user and root-correctness for the root-label user. The crossover p^* is the smallest p_w at which the mean Δ turns negative, by linear interpolation on the grid $p_w \in \{0.1, 0.2, \ldots, 0.9, 0.95\}$. A configuration is *safe* when Δ stays non-negative up to $p_w = 0.95$ and *robustly safe* when, in addition, the lower end of the crossover's bootstrap interval over orders is at 0.95, that is, no resample crosses earlier. Because Δ averages over all questions, we also measure it against what write-back could achieve. A question is *fixable* when static retrieval answers it wrongly and an earlier member of its family was answered correctly, so that a correct entry for it could exist. We report the share of fixable questions that write-back fixes, the share of statically correct questions it breaks, and the net change in correct answers per 1,000 questions. Intervals for Δ come from a joint bootstrap over orders and families, which accounts for the dependence between questions of the same family. Comparisons between conditions are paired by order, use two-sided paired t-tests, and are corrected for the twelve forum–retriever configurations with the Holm procedure.

Hypotheses. We organised the study around four hypotheses written down before the first simulation. They are stated here in their revised form; Appendix A gives the original wording and predictions and lists every way in which the analysis departs from that plan, including changes made after an internal code review. (H1) Entries written by the model displace the correct human thread more often than entries with the same content written by people. (H2) Errors propagate through provenance chains: entries generated from entries are served increasingly often, carry less of the original solution, and cause harm beyond that of first-level entries. (H3) A crossover estimate that accounts for depth predicts p^* on held-out orders better than the first-order estimate of Guthikonda (2026). (H4) Countermeasures differ in how much loss they remove at high p_w per unit of gain they give up at low p_w, and rules that act on provenance do better than rules that do not.

4 Results

4.1 How much is at stake

Write-back can only help a question whose family has already been answered correctly, so the room for improvement is small and differs widely between configurations: between 3% and 20% of questions are fixable (Table 1, Figure 2). With the question and generated answer design and the content-aware user, write-back fixes on average 3.7% of the fixable questions when users accept 10% of wrong answers and 3.5% when they accept 95%, and fixes almost none of the other statically wrong questions (0.1%). It breaks 0.5%

Table 1: How much is at stake, content-aware user, question and generated answer entries. Fixable: share of questions that static retrieval answers wrongly and that have an earlier, correctly answered member of their family. Fixed: share of fixable questions answered correctly with write-back. Broken: share of statically correct questions answered wrongly. Net: change in correct answers per 1,000 questions. Δ: points, 95% interval from a joint bootstrap over orders and families; the last column drops the largest family.

forum	retriever	static	fixable	fixed 0.1 / 0.95	other fixed 0.95	broken 0.95	net 0.95	Δ at 0.95	without largest
android	bge	15.6%	16.4%	4.2 / 3.9%	0.2%	3.4%	+1.5	+0.15 [-0.23, +0.57]	+0.17 [-0.30, +0.64]
android	MiniLM	20.3%	19.6%	3.5 / 3.0%	0.0%	2.2%	+1.4	+0.14 [-0.32, +0.49]	+0.06 [-0.40, +0.48]
android	BM25	12.6%	13.6%	1.2 / 1.4%	0.2%	1.1%	+1.8	+0.18 [-0.09, +0.48]	+0.21 [-0.08, +0.52]
unix	bge	19.8%	7.5%	6.8 / 6.9%	0.0%	1.7%	+2.1	+0.21 [-0.12, +0.54]	+0.23 [-0.10, +0.57]
unix	MiniLM	22.3%	8.3%	5.2 / 4.9%	0.2%	2.0%	+0.9	+0.09 [-0.33, +0.50]	+0.16 [-0.21, +0.51]
unix	BM25	13.4%	5.0%	2.9 / 2.7%	0.1%	1.7%	-0.1	-0.01 [-0.27, +0.25]	-0.01 [-0.26, +0.26]
webmasters	bge	10.5%	9.9%	2.6 / 2.1%	0.0%	3.7%	-1.9	-0.19 [-0.56, +0.12]	-0.23 [-0.67, +0.10]
webmasters	MiniLM	11.3%	17.1%	2.2 / 2.1%	0.1%	1.9%	+1.9	+0.19 [-0.15, +0.61]	+0.22 [-0.19, +0.68]
webmasters	BM25	8.7%	17.8%	0.8 / 0.9%	0.1%	0.6%	+1.6	+0.16 [-0.02, +0.41]	+0.19 [-0.03, +0.47]
mathematica	bge	11.6%	6.8%	1.1 / 0.8%	0.1%	2.5%	-1.7	-0.17 [-0.42, +0.06]	-0.18 [-0.46, +0.05]
mathematica	MiniLM	12.2%	4.2%	9.1 / 7.6%	0.1%	4.3%	-1.5	-0.15 [-0.53, +0.19]	-0.16 [-0.53, +0.21]
mathematica	BM25	9.0%	2.9%	5.2 / 5.3%	0.1%	3.2%	-0.7	-0.07 [-0.39, +0.23]	-0.08 [-0.37, +0.24]

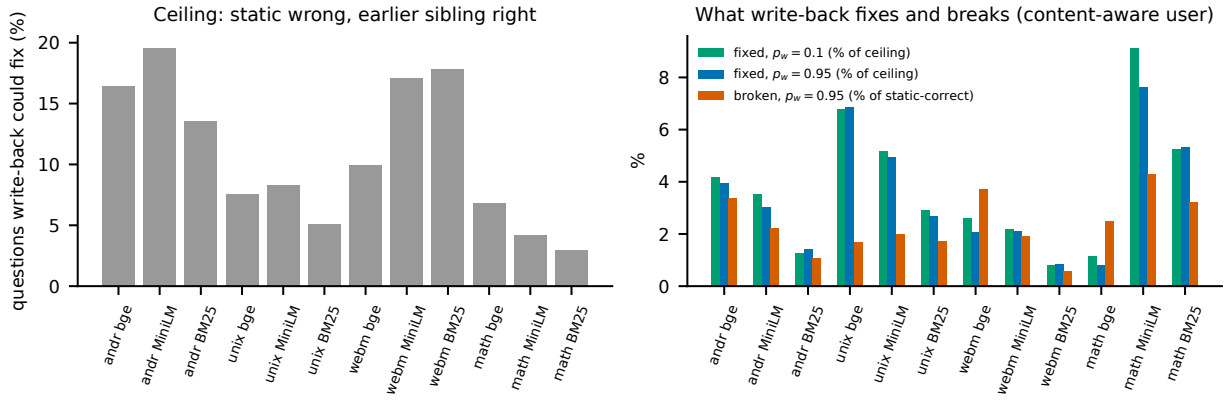


Figure 2: Left: share of questions that write-back could fix. Right: share of those it fixes at low and high acceptance of wrong answers, and share of statically correct questions it breaks at high acceptance. Content-aware user, question and generated answer entries.

and 2.3% of the questions static retrieval answered correctly. The net change ranges from +0.0 to +5.7 correct answers per 1,000 questions at $p_w = 0.1$ and from -1.9 to +2.1 at $p_w = 0.95$. The gain is significantly positive in 6 of 12 configurations at $p_w = 0.1$, and at $p_w = 0.95$ the interval includes zero in 12 of 12. The loop is not inert: in every configuration it both fixes and breaks answers. The net effect is small because few questions are fixable, write-back recovers only a few percent of them, and at high acceptance the fixes and breaks nearly cancel. Dropping the largest family, which holds up to 15% of a forum's questions, leaves every interval on the same side of zero (last column of Table 1).

4.2 H1: do retrievers prefer stored model-written answers?

Figure 3 shows Δ against p_w for every configuration. Entries that store the question with the generated answer are safe up to $p_w = 0.95$ in 7 of 12 configurations for the content-aware user, robustly so in 4. The question-only pointer entries of Guthikonda (2026) are safe in 3 and the question-with-thread pointer entries in 4. The median share of statically correct answers displaced by an entry at $p_w = 0.95$ is 12.7% for the question alone and about 3% for both question with thread (3.2%) and question with generated answer (3.1%). These comparisons differ in content as well as authorship.

The control holds the source thread and the length fixed (Table 2, Figure 4). When the entry is the answer alone, generated answers displace the correct thread more often than the human answers from the same

Table 2: Source-bias control, root-label user, $p_w = 0.95$. Each generated design is compared with the same entries carrying the root thread's human answer cut to the same length. Steal: correct static top-1 displaced by an entry; capture: an entry outscores the correct thread; both in %. Differences are generated minus human, averaged over the four forums. Last column: configurations where the paired difference in steal over orders is significant after Holm correction, positive / negative / total.

design pair	retriever	steal (gen.)	steal (human)	Δsteal	Δcapture	Δ (points)	sig. Δsteal
question + answer	bge	3.9	4.6	-0.7	-1.1	+0.08	0/1/4
question + answer	MiniLM	3.8	3.1	+0.7	+0.0	+0.04	0/0/4
question + answer	BM25	2.7	1.4	+1.3	+2.9	-0.01	2/0/4
answer only	bge	8.6	7.1	+1.5	+3.0	+0.03	1/0/4
answer only	MiniLM	3.0	1.8	+1.2	+3.1	+0.09	2/0/4
answer only	BM25	2.4	1.3	+1.0	+4.4	+0.07	2/0/4

thread: the steal rate is higher in 10 of 12 configurations, significantly so in 5 after Holm correction and significantly lower in none, by +1.5 points for bge, +1.2 for MiniLM and +1.0 for BM25. This is the direction of the source bias reported for static corpora (Dai et al., 2024b). That it appears for BM25 as well suggests that part of it is lexical: the generated answer was written in response to its question and restates its terms, which also favours it for the next, similar question (Huang et al., 2025). When the question is stored with the answer, the difference changes to -0.7 points for bge and +0.7 for MiniLM but stays at +1.3 for BM25, and it is significant in 2 configurations in one direction and 1 in the other. The consequence for accuracy is small: averaged over forums, the paired difference in Δ at $p_w = 0.95$ is at most a tenth of a point, of the same size as the effect of write-back itself at that acceptance rate, and of mixed sign across configurations (Figure 4, right). H1 holds in direction for answer-only entries and weakly for question and answer entries, and the control cannot separate authorship from question conditioning.

4.3 H2: do errors propagate along provenance chains?

Chains form. At $p_w = 0.95$, 5% to 13% of entries were generated from another entry, and the share of answers served from such entries is still rising at the end of the stream, reaching 4.9% in the last round (Figure 5, left). The text degrades along the chain. The share of served answers in which the judge finds the root thread's solution is lower for answers served from depth-2 or deeper entries than for answers generated from a thread in 12 of 12 configurations (Wilcoxon $p = 0.0005$), 58% against 76% pooled over served answers, with depth-1 entries in between at 62% (Figure 5, middle). Most of the change is from a full to a partial solution rather than to a wrong one. The three-judge audit of independently sampled answers shows the same trend (Appendix C), and similarity to the human answer falls from 0.85 to 0.82 and 0.80 by depth, where an unrelated human answer scores about 0.60.

Measurable harm does not follow within the stream. Relative to static retrieval, the answers served from depth-1 entries change the number of correct answers by +0.37 per 1,000 questions on average over configurations (95% interval -0.51 to +1.23), from depth-2 entries by +0.06 (-0.19 to +0.30) and from deeper entries by -0.01 (-0.11 to +0.07; Figure 5, right). Every interval includes zero. Wrong answers served from depth-2 entries are 0.3% to 2.5% of all wrong answers, against 4.6% to 11.6% for depth 1, so deep entries are both rare and, on average, no worse than static retrieval for the questions they serve. These contrasts are against static retrieval, not against the shallower entry a deep answer descends from, and the stream is a single pass through each forum's questions. H2 is supported for the text and for the growth of chains; we find no evidence of harm from depth within the stream, and the horizon check in Section 4.6 finds none over twice the length.

4.4 H3: can the crossover be predicted?

The first-order estimate of Guthikonda (2026) predicts p^* from the static accuracy and the average effects of a correct and a wrong entry measured at one reference p_w. Our depth-aware estimate keeps the per-entry effects separate for depth-1 and deeper wrong entries and lets the share of deep wrong entries grow

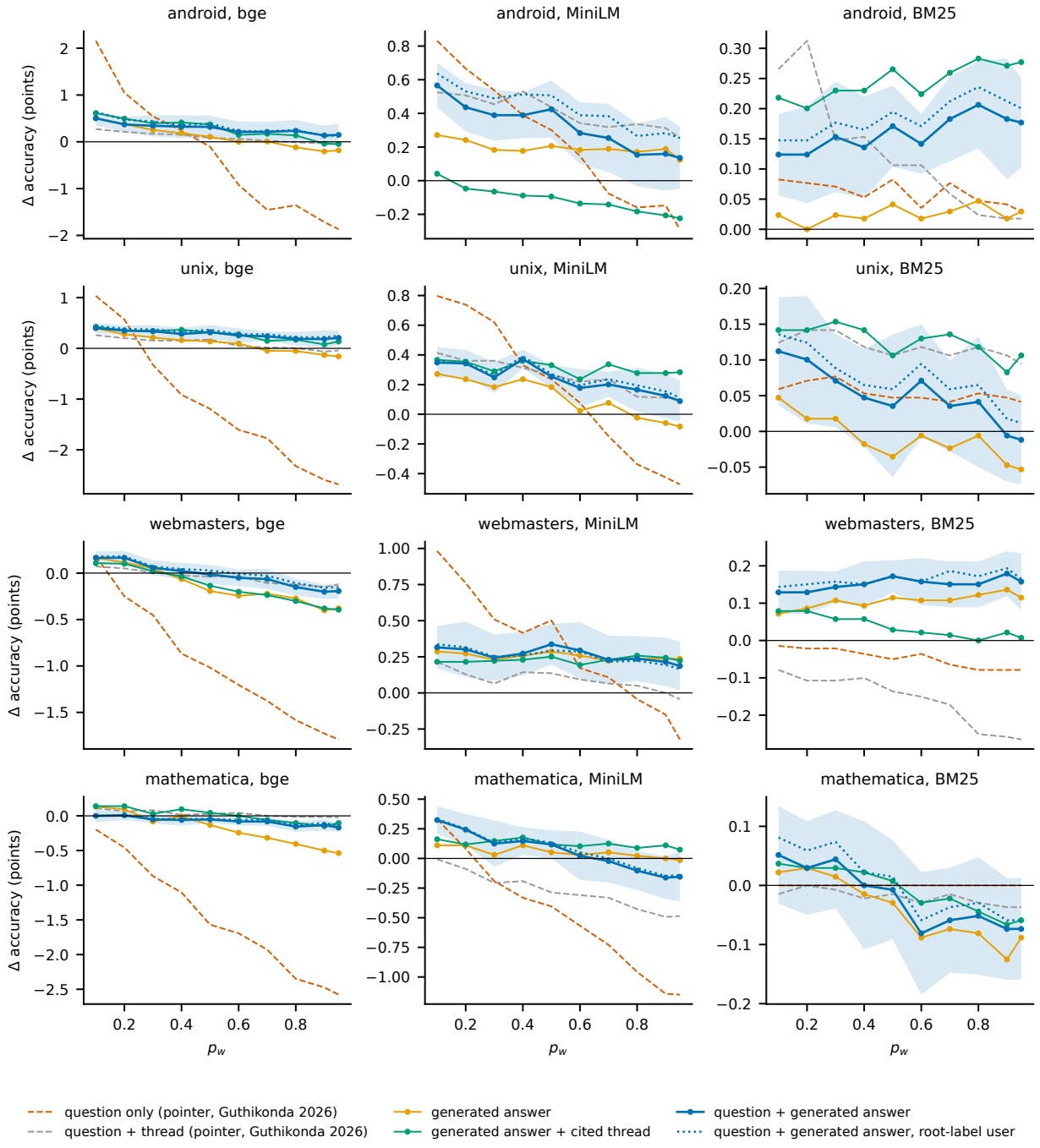


Figure 3: Δ against the acceptance rate of wrong answers, by forum and retriever. Solid: generated-answer entry designs, content-aware user, with a 95% band over orders for question and generated answer. Dotted: the same design with the root-label user. Dashed: pointer designs of Guthikonda (2026), root label.

in proportion to p_w, since each further level needs one more wrong acceptance. At the reference point both estimates coincide; they differ only in how they extrapolate. Table 3 and Figure 6 evaluate both on held-out orders. Few configurations have a crossover inside the grid in both halves of the orders. For question and generated answer entries at $p_{\text{ref}} = 0.3$ there are 6 configuration–split pairs, and on them the first-order estimate has a mean absolute error of 0.249 and the depth-aware estimate 0.207 (Wilcoxon $p = 0.12$), larger

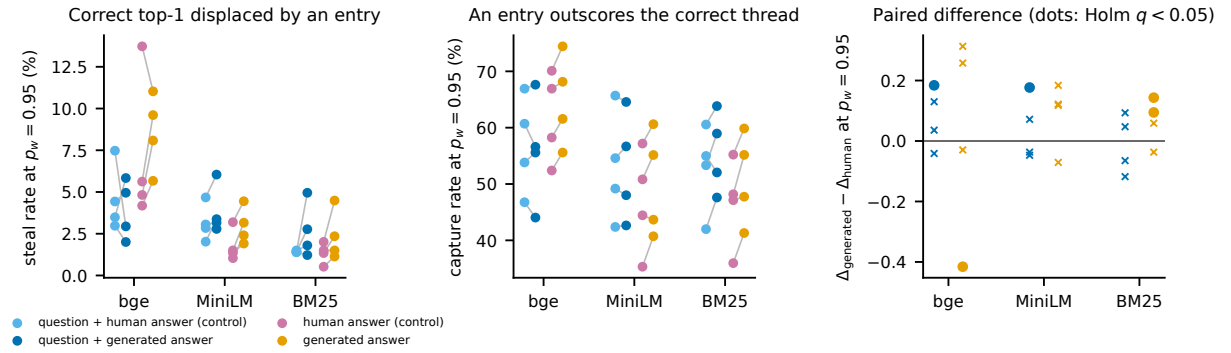


Figure 4: Source-bias control at $p_w = 0.95$, root-label user. Each line joins a human control (left point) and its generated design (right point) in one forum. Right panel: paired difference in Δ per configuration; filled dots are significant after Holm correction.

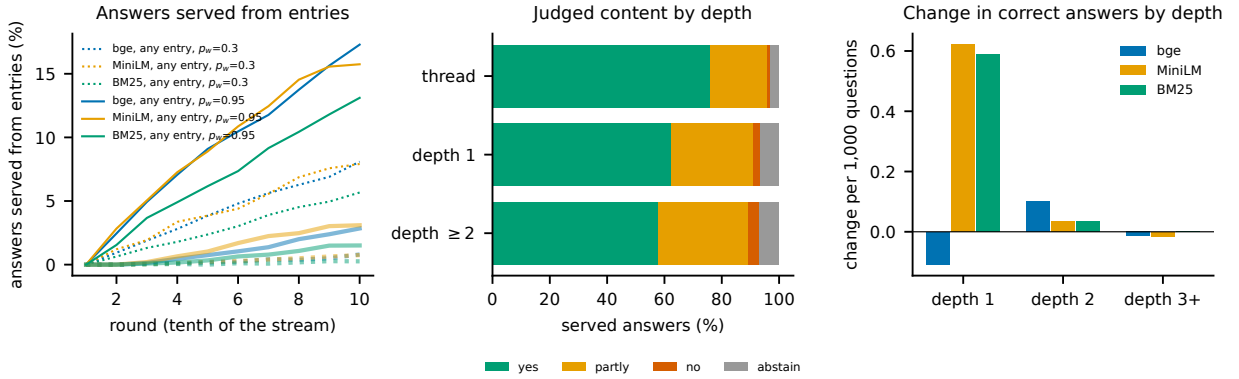


Figure 5: Depth, content-aware user, question and generated answer entries. Left: share of answers served from any entry (thin) and from an entry of depth ≥ 2 (thick), mean over forums, at $p_w = 0.3$ (dotted) and 0.95 (solid). Middle: judged content of served answers by the depth of their evidence, pooled, $p_w = 0.95$. Right: change in correct answers per 1,000 questions attributable to answers served at each depth, relative to static retrieval, $p_w = 0.95$.

than the 0.097 obtained by predicting every observed crossover by their mean, a reference that uses the held-out values and is therefore optimistic. For answer-only entries both estimates do better than that reference (Table 3). The depth-aware estimate is never significantly better than the first-order one. H3 is not supported, and for the main design neither estimate is reliable.

4.5 H4: which countermeasures work?

Table 4 and Figure 7 compare the rules with the content-aware user. The oracle reopen used by Guthikonda (2026) is naive write-back at half the acceptance rate: its Δ curve matches naive write-back at $p_w/2$ within 0.03 points on average. The realistic forms keep part of its benefit. Reopening after one round raises the mean Δ at $p_w = 0.95$ from +0.04 to +0.14 points and costs nothing at $p_w = 0.1$ (+0.26 against +0.25); it is significantly better than naive write-back at $p_w = 0.95$ in 3 of 12 configurations and worse in 0. Waiting two rounds gives +0.10, and reopening 10% or 20% of correct entries by mistake gives +0.11 and +0.09. Neither the depth cap (+0.04) nor regenerating from the root thread (+0.05) improves on naive write-back, consistent with H2: when deep entries carry no measurable harm, removing or repairing them changes little. Down-weighting, uniform or by depth, makes write-back nearly inert: it removes the gain at $p_w = 0.1$ in most configurations (significantly worse than naive in 8 of 12 for the depth penalty with $\lambda = 0.5$), and its

Table 3: Predicting the crossover on held-out orders (calibrate on five orders, evaluate on the other five, both directions; root-label user). MAE over configuration–split pairs where both the observed and the predicted crossover lie inside the grid, with a bootstrap 95% interval. Mean baseline: predicting every observed crossover by their mean. Paired: the two estimates on the pairs where both are defined, with a Wilcoxon signed-rank test.

design	p_{ref}	n	first-order MAE	depth-aware MAE	mean baseline	Spearman (first-order)	Wilcoxon p
question + answer	0.3	6	0.249 [0.155, 0.340]	0.207 [0.114, 0.306]	0.097	-0.43	0.12
question + answer	0.5	6	0.152 [0.081, 0.237]	0.137 [0.056, 0.228]	0.097	0.03	0.09
answer only	0.3	14	0.203 [0.130, 0.281]	0.198 [0.115, 0.291]	0.213	0.33	0.86
answer only	0.5	13	0.165 [0.098, 0.254]	0.157 [0.098, 0.230]	0.221	0.64	0.76
answer + thread	0.3	7	0.217 [0.164, 0.269]	0.223 [0.132, 0.314]	0.155	-0.14	0.08
answer + thread	0.5	6	0.133 [0.039, 0.244]	0.139 [0.045, 0.240]	0.151	0.43	0.16

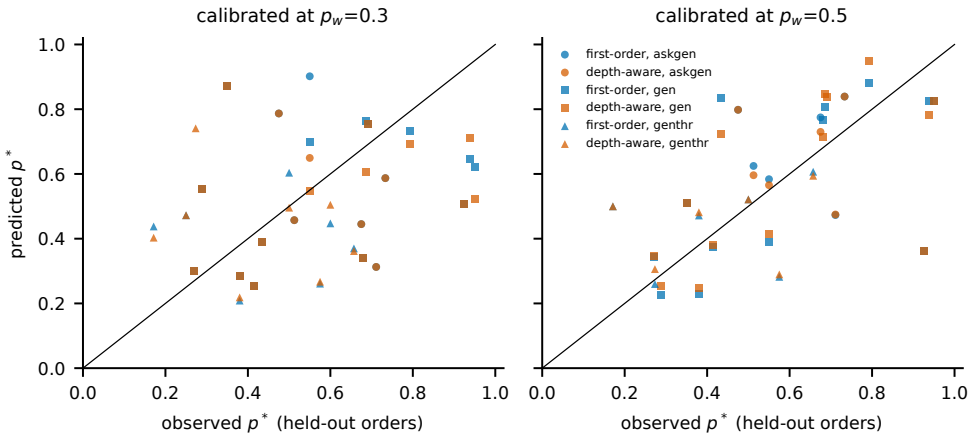


Figure 6: Predicted against observed crossover on held-out orders, for the three generated designs.

higher count of safe configurations reflects that little is left to go wrong. H4 is supported in that the rules differ sharply in what they give up for what they protect. Its second part is not: no rule that acts on provenance improves on delayed, noisy reopen, although no rule is significantly better than naive write-back in more than a few configurations.

4.6 Robustness

User model. Judging what the answer says rather than where it came from lowers the benefit slightly, from +0.29 to +0.25 points at $p_w = 0.1$ and from +0.07 to +0.04 at $p_w = 0.95$, and moves the crossover earlier in 4 configurations (Figure 3, dotted against solid). The source-bias control and the crossover estimates use the root-label user, because both concern retrieval rather than acceptance; Appendix D gives every design for both users.

Prompt: instruction or length? A first version of this check changed two things at once. We therefore ran all four combinations of the instruction (answer from the evidence only, or answer as helpfully as possible using the evidence as context) and the length limit (120 words, or none) on all four forums (Table 5, Figure 8). The direction points to the instruction. Averaged over the length conditions and paired over the twelve configurations, dropping the evidence-only instruction widens the fall in the share of answers that state the root solution, from a thread to a depth-2 entry, by +14.1 points, in 10 of 12 configurations and in all four forums, though strongly in only two of them. Removing the length limit changes that fall by -0.1 points. After Holm correction over the ten tests in Table 5, however, the instruction effect is not significant ($q = 0.110$), and the only effect that survives is that removing the length limit raises the share of correct answers generated directly from a thread by +3.4 points ($q = 0.005$). The prompt changes neither Δ nor

Table 4: Countermeasures, content-aware user, question and generated answer entries, mean over the twelve configurations. AUC: mean Δ over the p_w grid. Safe / robust: configurations with no crossover below 0.95, and those where in addition no bootstrap resample of the orders crosses earlier. Better / worse: configurations where the rule is significantly better or worse than naive write-back at $p_w = 0.95$, and significantly worse at $p_w = 0.1$ (paired over orders, Holm-corrected).

rule	Δ at 0.1	Δ at 0.95	AUC	safe / robust	better / worse at 0.95	worse at 0.1
naive write-back	+0.25	+0.04	+0.14	7 / 4	–	–
reopen, oracle (Guthikonda, 2026)	+0.26	+0.15	+0.20	9 / 7	5 / 0	0
reopen after 1 round	+0.26	+0.14	+0.19	9 / 7	3 / 0	0
reopen after 2 rounds	+0.26	+0.10	+0.18	8 / 7	3 / 0	0
reopen after 1 round, 10% false	+0.24	+0.11	+0.16	8 / 6	2 / 0	0
reopen after 1 round, 20% false	+0.22	+0.09	+0.14	8 / 5	2 / 0	0
depth cap (depth 1 only)	+0.24	+0.04	+0.13	7 / 5	0 / 0	1
regenerate from root	+0.25	+0.05	+0.14	7 / 6	0 / 1	0
uniform penalty, $\lambda = 0.5$	+0.06	+0.02	+0.04	8 / 4	2 / 2	8
uniform penalty, $\lambda = 1$	+0.01	+0.01	+0.01	10 / 9	2 / 2	9
depth penalty, $\lambda = 0.5$	+0.06	+0.03	+0.04	8 / 5	3 / 2	8
depth penalty, $\lambda = 1$	+0.01	+0.01	+0.01	10 / 9	2 / 2	9

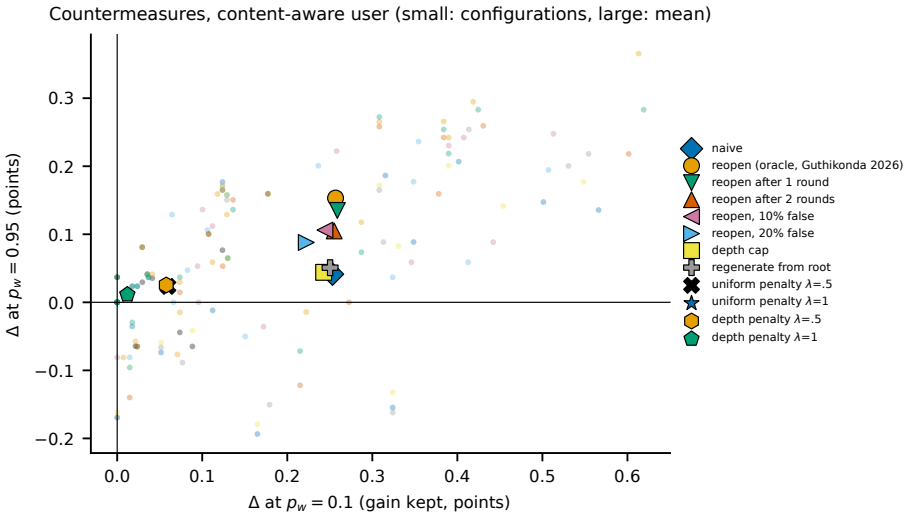


Figure 7: Gain kept at $p_w = 0.1$ against the outcome at $p_w = 0.95$, per configuration (small) and mean (large), content-aware user.

the estimated change from depth-2 entries detectably, and every cell's harm interval includes zero. The evidence-only instruction is therefore the likely reason drift stays moderate in the main results, but our data cannot establish it firmly; with only four forums, the twelve configurations are not independent.

Generator. Figure 9 repeats the main design with an open-weight generator on android. It states the root solution less often at every depth (52% from threads, 46% from depth-2 entries) and is safe in 2 of 3 configurations; its depth effects have wide intervals.

Horizon. Each forum contributes one pass of real questions, and the share of answers served from deep entries is still rising at its end. To give chains more room, we added a paraphrase of every question, generated once by `gpt-5.4-mini` and assigned to the same family, and ran the main design over twenty rounds of the same size, with originals and paraphrases in one random order. The original questions' rankings are unchanged. Deep entries now become common: at $p_w = 0.95$ the share of answers served from depth-2 or deeper entries grows from 6.4% half-way to 15.8% in the last round (up to 24.6%; Figure 10), and 35% to

Table 5: Prompt factorial, content-aware user, question and generated answer entries, all twelve configurations. Yes: share of served answers that state the root thread's solution, from a thread and from an entry of depth 2 or more; drop: their difference in points (mean over configurations). Harm: change in correct answers per 1,000 questions from answers served at depth 2 or more, with a joint bootstrap 95% interval.

instruction	length	yes, thread	yes, depth ≥ 2	drop	Δ at 0.1	Δ at 0.95	safe	harm at depth 2
evidence-only	120 words	76%	54%	21.7	+0.25	+0.04	7/12	+0.06 [-0.19, +0.30]
evidence-only	no limit	79%	61%	17.9	+0.22	-0.01	6/12	-0.07 [-0.32, +0.14]
open	120 words	79%	47%	32.1	+0.16	-0.06	3/12	-0.12 [-0.34, +0.10]
open	no limit	83%	47%	35.6	+0.20	-0.03	5/12	-0.12 [-0.34, +0.09]
instruction (open − evidence-only)		+3.2 (q=0.110)	-10.9 (q=0.297)	+14.1 (q=0.110)	-0.06 (q=0.320)	-0.06 (q=0.297)		
length (no limit − 120 words)		+3.4 (q=0.005)	+3.5 (q=0.605)	-0.1 (q=1.000)	+0.01 (q=1.000)	-0.01 (q=1.000)		

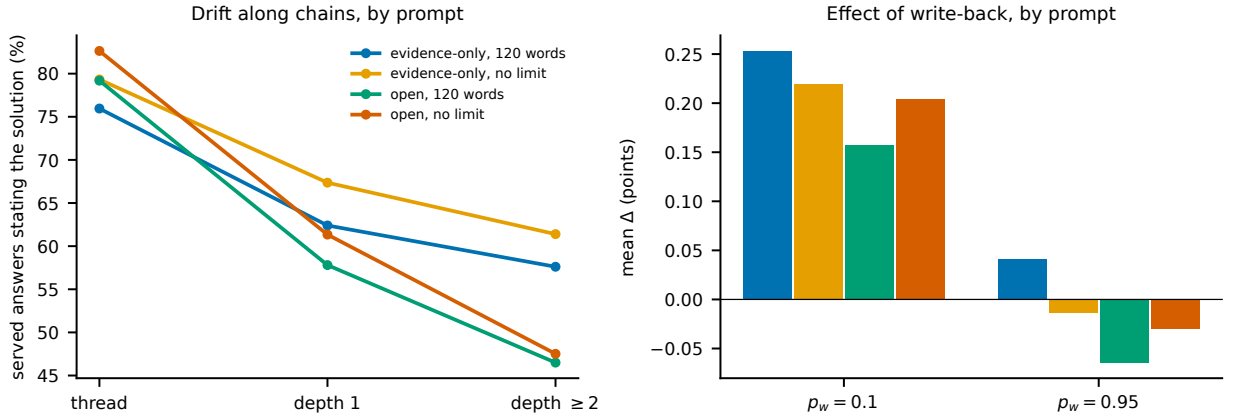


Figure 8: Prompt factorial, all four forums, content-aware user. Left: share of served answers that state the root solution, by evidence depth. Right: mean Δ at low and high acceptance of wrong answers.

49% of entries are deep. The text still degrades with depth, in 12 of 12 configurations. The pooled change in correct answers attributable to depth-2 entries is +0.05 per 1,000 questions (-0.49 to +0.57) and to deeper entries +0.05 (-0.26 to +0.38), against -0.29 for depth-1 entries (-1.21 to +0.64); one of 36 per-configuration intervals excludes zero. So even when deep entries are common we detect no pooled harm from them. Reopening after one round raises the mean Δ at $p_w = 0.95$ from -0.02 to +0.52 points and is significantly better than naive write-back in 12 of twelve configurations. Three features of this design limit how far it generalises. Paraphrases are easier to match than real duplicates, which inflates the benefit of write-back and of reopen. A paraphrase is often served from its own twin's entry, whose static answer tends to share its correctness, which pulls the harm estimate towards zero. And the same model writes the paraphrases and the answers, which may strengthen the match between stored entries and new questions. The check shows that deep entries need not be harmful when common; it is not a simulation of a system running for years.

Largest family. Removing the largest family leaves every interval in Table 1 on the same side of zero.

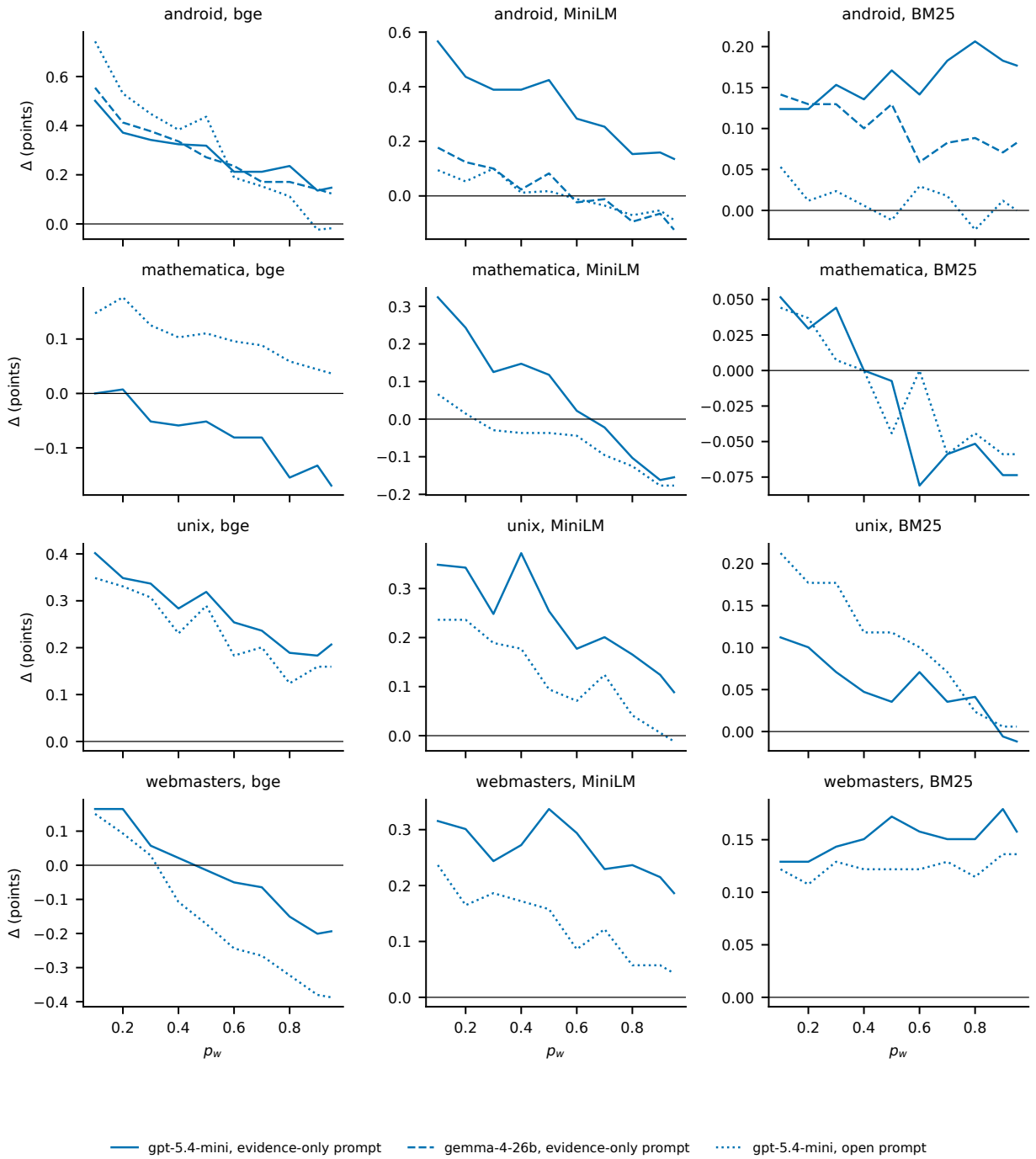


Figure 9: Δ for question and generated answer entries, content-aware user: main generator and prompt, open-weight generator (android), and the open prompt without the evidence-only instruction or length limit.

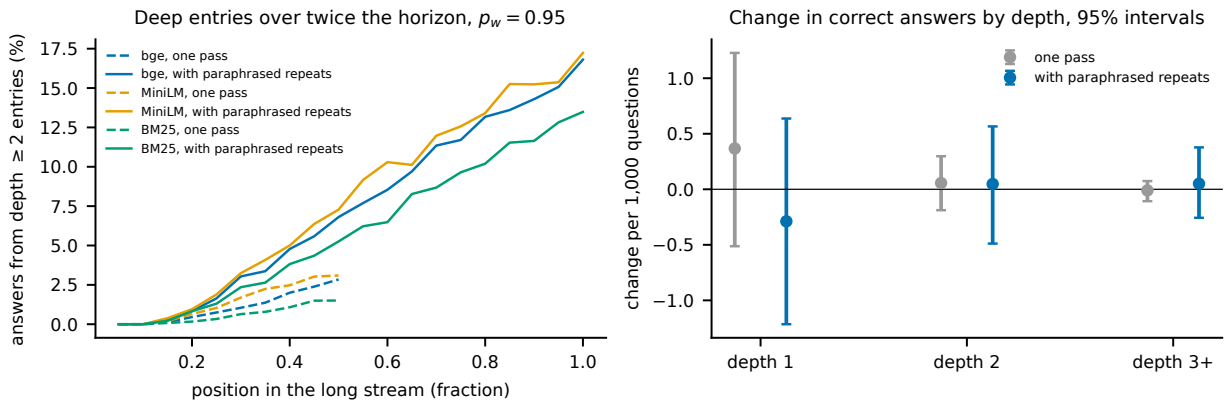


Figure 10: Long stream with paraphrased repeats, content-aware user, question and generated answer entries, $p_w = 0.95$. Left: share of answers served from entries of depth ≥ 2 over the doubled stream, against the one-pass stream (dashed, plotted over the first half for scale only, since the two streams are ordered differently). Right: change in correct answers per 1,000 questions by served depth with joint bootstrap 95% intervals.

5 Discussion and limitations

What a practitioner should take from this. On forums like these, writing accepted generated answers back is a small lever in both directions: most questions cannot benefit, those that can are fixed only a few percent of the time, and at high acceptance of wrong answers the fixes and breaks cancel. Within the horizon we observe, the risk worth managing is the first-level one, a wrongly accepted answer retrieved for the next similar question, rather than a chain. Store the question with the answer, which reduces the dense retrievers' preference for stored generated text. Instruct the generator to answer from the retrieved evidence: without that instruction stored text tended to drift faster along chains in our factorial, though not significantly after correction, while limiting answer length did not matter. Let users reopen answers that did not work and remove the corresponding entries; this keeps the gain even when reopening is late or sometimes wrong, though it helps significantly in only some configurations. Rules that act on provenance depth do not improve on it, and down-weighting stored entries buys safety only by switching the loop off.

Limitations. Acceptance is simulated with a constant p_w, while real users probably accept fluent wrong answers more often than hesitant ones; the content-aware user captures part of this, not all. The content label comes from one judge model, validated against two others with moderate agreement (κ between 0.48 and 0.64); no answers were labelled by people, and judges can favour model-written text (Panickssery et al., 2024). A PARTLY verdict counts as not correct, and most of the degradation with depth is a shift from YES to PARTLY. Correctness is anchored on the root thread's human answer, which ignores answers that are right from the model's own knowledge despite wrong evidence; this matters more for the open prompt. The source-bias control holds the root and length fixed but not question conditioning. The streams are one pass through the held-out questions of each forum, doubled in the horizon check by paraphrased repeats rather than new real questions; deep entries keep growing over the doubled stream, so a system that runs for years could still see compounding we cannot measure, and our harm-by-depth estimates are relative to static retrieval. The questions are CQADupStack titles, the four forums are technical, and the static accuracy of the retrievers is low, which limits how much any entry can help. Answers were fetched in 2026 for a 2014 snapshot, so some accepted answers have been edited since.

6 Conclusion

When generated answers become evidence in a human-gated support loop, the text drifts along provenance chains in every configuration we measured and retrievers show a small preference for model-written entries. Over one pass of real repeat questions, and in a longer stream with paraphrased repeats where deep entries become common, neither translates into detectable harm, and the effect of write-back is small against what it could achieve. The most useful protections are simple ones: instruct the generator to answer from its evidence, store the question with the answer, and let users reopen what did not work, which helped in every configuration of the longer stream.

Broader Impact Statement

This work studies a design that support and service desks already deploy: storing answers that users accepted so that later questions can be answered from them. The results are meant to help practitioners decide whether and how to do this safely. The data are public community question-and-answer threads, used under their CC BY-SA licence, and no personal or customer data were used. The simulated user is a simplification, and a deployment that relies on our numbers without measuring its own acceptance of wrong answers could still store harmful content. Generated answers are stored only after acceptance in our setting; systems that write back without any human or automated check are outside what we measure.

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A Deviations from the initial plan

The study plan was written before the first simulation and is included in the supplementary material. Table 6 lists every departure from it. The most important ones followed an internal review of the first version of the code and the paper: the first version labelled correctness only by the root thread, so the generated text could not create an error; its source-bias test did not hold content fixed; its depth-aware estimate was, by construction, the first-order estimate divided by a constant; and its reopen rule was an oracle. All four are corrected here and the first-version numbers are not used.

Original hypotheses and predictions. The plan stated H1 as "at equal acceptance, stored generated answers win top-1 more often than stored thread entries" and predicted that the crossover for generated entries would lie between those of the two pointer designs; generated entries were instead safer than both. It stated H2 as "the share of wrong answers whose evidence is itself a wrong stored generated answer grows with rounds". It stated H3 as "the first-order estimate under-predicts the loss when H2 holds; a second-order correction restores the fit". It predicted for H4 that reopen would still win and that provenance down-weighting would be the cheapest rule; reopen did best, and down-weighting removed the gain. The revised wording in Section 3 adds the content-aware user and the same-source control.

B Prompts

Generator, main prompt. *You are a technical support assistant. Answer the user's question using only the evidence provided. If the evidence does not fully answer the question, say what it does cover and give the closest guidance it supports. At most 120 words. Plain text, no headings, no mention of the evidence or of these instructions.* The user message is the question followed by the evidence. A thread is shown as its text followed by its accepted or top-voted answer, or a note that it has none. A stored entry is shown as "Previously resolved case" with its question and stored answer.

Generator, open prompt. *You are a technical support assistant. Answer the user's question as helpfully as you can. A previously resolved case or thread is provided as context; use it where it helps. Plain text.* No length limit beyond the token cap.

Table 6: Departures from the initial plan.

Item	Planned	Done, and why
H1	Compare crossover and capture of generated entries with the pointer entries of Guthikonda (2026).	Kept, and added a same-source control with length-matched human text, because the planned comparison confounds authorship with content and length.
H2	Compounding shown by deep entries growing with rounds and by drift.	Kept, and added content labels for every served answer and counterfactual harm by served depth, because growth alone does not show harm.
H3	Depth-aware plug-in estimate, tested out of sample.	The first version used an estimator that only rescaled the first-order one. Replaced by the estimator of Section 4.4, tested out of sample as planned.
H4	Rank reopen, cited source and down-weighting by loss removed per gain lost.	Reopen in the plan was the oracle rule of Guthikonda (2026). Added delayed and noisy reopen, a depth cap, regeneration from the root and depth-proportional down-weighting. Storing the cited source is reported under H1, where it is an entry design.
Correctness	Evidence-based root label, plus a 200-answer manual audit.	Root label kept as one of two user models. The content label from an independent judge replaces the manual audit as the main label, validated against two further judges on 900 answers (Appendix C). No answers were judged by hand.
Data	Four CQADupStack forums and MS MARCO query families.	MS MARCO dropped: its passages have no community answer to judge content against.
Generators	One generator, one open-weight replication.	Kept, and added a full factorial over the prompt, instruction by length limit, on all four forums (Section 4.6).
Horizon	One pass per forum.	Added a doubled stream with paraphrased repeats on all four forums (Section 4.6).
Human labels	200-answer manual audit.	Not done; validation is against two further model judges only.

Judge. *You grade answers from a technical support system. You are given the user's question, a reference answer written by the community and accepted as the solution, and the system's answer. Reply with exactly one word. YES: the system's answer gives the same solution as the reference, possibly with less detail. PARTLY: it overlaps with the reference but misses or changes the key step. NO: it gives a different solution, a wrong solution, or no usable solution. ABSTAIN: it says the information available does not answer the question and does not offer a solution.* The reference is the root thread's human answer, cut to 600 words.

C Judge validation and faithfulness audit

D Full results

Table 7: Faithfulness audit of 900 cached answers, 25 per forum at each evidence depth and order of sampling, judged by three models against the root thread's human answer (ABSTAIN counted as NO). Cohen's κ: DeepSeek and gpt-5.4-mini 0.64, DeepSeek and gemma 0.48, gpt-5.4-mini and gemma 0.52. Trend p: Cochran–Armitage test for the share of NO across depths.

	DeepSeek-V4.1-Flash			gpt-5.4-mini			gemma-4-26b		
evidence depth	yes	partly	no	yes	partly	no	yes	partly	no
0 (thread)	73%	20%	7%	73%	17%	10%	86%	3%	12%
1	61%	33%	7%	61%	28%	11%	78%	7%	15%
2 or more	55%	31%	14%	57%	27%	16%	73%	8%	19%
trend p (no)			0.003			0.025			0.009

Table 8: Entry designs by user model and configuration: crossover p^* with bootstrap interval, Δ (points, mean $\pm$ 95% interval over orders) at $p_w = 0.1$ and 0.95, share of answers served from entries and steal rate at $p_w = 0.95$. Pointer rows are from Guthikonda (2026) (root label).

config	user	design	p^*	[95% interval]	Δ at 0.1	Δ at 0.95	self	steal
android bge	root	question only (pointer)	0.47	[0.37, 0.54]	+2.16 ± 0.72	-1.87 ± 0.65	65%	33.5%
android bge	root	question + thread (pointer)	0.77	[0.49, 0.95]	+0.27 ± 0.07	-0.04 ± 0.14	19%	3.5%
android bge	root	answer	0.74	[0.57, 0.83]	+0.77 ± 0.12	-0.19 ± 0.21	27%	9.6%
android bge	root	question + answer	>0.95	[0.83, 0.95]	+0.64 ± 0.15	+0.14 ± 0.24	19%	5.8%
android bge	root	answer + thread	0.89	[0.81, 0.95]	+0.81 ± 0.12	-0.04 ± 0.21	25%	8.7%
android bge	root	human answer (ctrl)	0.60	[0.34, 0.74]	+0.31 ± 0.07	-0.16 ± 0.13	19%	5.6%
android bge	root	question + human answer (ctrl)	>0.95	[0.85, 0.95]	+0.37 ± 0.10	+0.18 ± 0.16	15%	3.5%
android bge	content	answer	0.60	[0.46, 0.83]	+0.51 ± 0.13	-0.18 ± 0.23	28%	9.6%
android bge	content	question + answer	>0.95	[0.84, 0.95]	+0.50 ± 0.15	+0.15 ± 0.24	19%	5.8%
android bge	content	answer + thread	0.88	[0.60, 0.95]	+0.61 ± 0.14	-0.05 ± 0.19	25%	8.7%
android bge	content	human answer (ctrl)	0.60	[0.29, 0.75]	+0.27 ± 0.09	-0.14 ± 0.11	19%	5.6%
android bge	content	question + human answer (ctrl)	>0.95	[0.74, 0.95]	+0.28 ± 0.09	+0.10 ± 0.14	15%	3.5%
android MiniLM	root	question only (pointer)	0.67	[0.55, 0.78]	+0.83 ± 0.31	-0.29 ± 0.24	34%	9.6%
android MiniLM	root	question + thread (pointer)	>0.95	[0.95, 0.95]	+0.52 ± 0.19	+0.25 ± 0.14	24%	3.0%
android MiniLM	root	answer	>0.95	[0.95, 0.95]	+0.33 ± 0.10	+0.15 ± 0.12	10%	2.4%
android MiniLM	root	question + answer	>0.95	[0.95, 0.95]	+0.64 ± 0.14	+0.25 ± 0.17	15%	2.8%
android MiniLM	root	answer + thread	0.38	[0.20, 0.75]	+0.12 ± 0.06	-0.14 ± 0.14	9%	2.6%
android MiniLM	root	human answer (ctrl)	>0.95	[0.19, 0.95]	+0.11 ± 0.07	+0.04 ± 0.08	8%	1.0%
android MiniLM	root	question + human answer (ctrl)	>0.95	[0.84, 0.95]	+0.28 ± 0.07	+0.08 ± 0.14	16%	2.8%
android MiniLM	content	answer	>0.95	[0.95, 0.95]	+0.27 ± 0.09	+0.12 ± 0.11	10%	2.4%
android MiniLM	content	question + answer	>0.95	[0.87, 0.95]	+0.57 ± 0.13	+0.14 ± 0.18	15%	2.8%
android MiniLM	content	answer + thread	0.15	[0.10, 0.22]	+0.04 ± 0.08	-0.22 ± 0.11	9%	2.6%
android MiniLM	content	human answer (ctrl)	0.75	[0.10, 0.95]	+0.05 ± 0.07	+0.01 ± 0.07	8%	1.0%
android MiniLM	content	question + human answer (ctrl)	>0.95	[0.58, 0.95]	+0.22 ± 0.09	+0.01 ± 0.13	16%	2.8%
android BM25	root	question only (pointer)	>0.95	[0.34, 0.95]	+0.08 ± 0.05	+0.03 ± 0.10	7%	1.8%
android BM25	root	question + thread (pointer)	>0.95	[0.46, 0.95]	+0.27 ± 0.18	+0.02 ± 0.19	19%	4.5%
android BM25	root	answer	>0.95	[0.17, 0.95]	+0.05 ± 0.04	+0.05 ± 0.07	6%	1.1%
android BM25	root	question + answer	>0.95	[0.95, 0.95]	+0.15 ± 0.07	+0.20 ± 0.08	10%	1.2%
android BM25	root	answer + thread	>0.95	[0.95, 0.95]	+0.24 ± 0.08	+0.29 ± 0.11	9%	0.3%
android BM25	root	human answer (ctrl)	<0.10	[0.10, 0.10]	-0.04 ± 0.02	-0.01 ± 0.03	3%	0.5%
android BM25	root	question + human answer (ctrl)	>0.95	[0.95, 0.95]	+0.22 ± 0.08	+0.27 ± 0.09	10%	1.4%
android BM25	content	answer	>0.95	[0.13, 0.95]	+0.02 ± 0.03	+0.03 ± 0.06	6%	1.1%
android BM25	content	question + answer	>0.95	[0.95, 0.95]	+0.12 ± 0.07	+0.18 ± 0.07	10%	1.2%
android BM25	content	answer + thread	>0.95	[0.95, 0.95]	+0.22 ± 0.07	+0.28 ± 0.08	9%	0.3%
android BM25	content	human answer (ctrl)	<0.10	[0.10, 0.10]	-0.04 ± 0.02	-0.01 ± 0.03	3%	0.5%
android BM25	content	question + human answer (ctrl)	>0.95	[0.95, 0.95]	+0.20 ± 0.07	+0.24 ± 0.09	10%	1.4%
unix bge	root	question only (pointer)	0.26	[0.24, 0.29]	+1.03 ± 0.38	-2.68 ± 0.57	56%	29.9%
unix bge	root	question + thread (pointer)	0.81	[0.58, 0.95]	+0.26 ± 0.08	-0.04 ± 0.14	11%	2.7%
unix bge	root	answer	0.83	[0.61, 0.95]	+0.45 ± 0.13	-0.08 ± 0.20	16%	5.7%
unix bge	root	question + answer	>0.95	[0.95, 0.95]	+0.44 ± 0.08	+0.25 ± 0.15	8%	2.0%
unix bge	root	answer + thread	>0.95	[0.95, 0.95]	+0.48 ± 0.07	+0.22 ± 0.13	12%	3.2%
unix bge	root	human answer (ctrl)	<0.10	[0.10, 0.10]	-0.04 ± 0.05	-0.40 ± 0.10	13%	4.2%
unix bge	root	question + human answer (ctrl)	>0.95	[0.95, 0.95]	+0.38 ± 0.07	+0.12 ± 0.12	10%	3.0%
unix bge	content	answer	0.67	[0.53, 0.95]	+0.40 ± 0.14	-0.16 ± 0.20	16%	5.7%
unix bge	content	question + answer	>0.95	[0.95, 0.95]	+0.40 ± 0.10	+0.21 ± 0.14	8%	2.0%
unix bge	content	answer + thread	>0.95	[0.88, 0.95]	+0.43 ± 0.09	+0.14 ± 0.11	12%	3.2%
unix bge	content	human answer (ctrl)	<0.10	[0.10, 0.10]	-0.04 ± 0.05	-0.40 ± 0.10	13%	4.2%
unix bge	content	question + human answer (ctrl)	>0.95	[0.95, 0.95]	+0.38 ± 0.07	+0.12 ± 0.12	10%	3.0%
unix MiniLM	root	question only (pointer)	0.63	[0.52, 0.74]	+0.80 ± 0.17	-0.47 ± 0.24	27%	11.2%
unix MiniLM	root	question + thread (pointer)	>0.95	[0.84, 0.95]	+0.41 ± 0.14	+0.10 ± 0.16	16%	3.0%
unix MiniLM	root	answer	0.84	[0.59, 0.95]	+0.27 ± 0.08	-0.05 ± 0.12	8%	3.2%
unix MiniLM	root	question + answer	>0.95	[0.95, 0.95]	+0.35 ± 0.10	+0.11 ± 0.13	10%	3.2%
unix MiniLM	root	answer + thread	>0.95	[0.95, 0.95]	+0.37 ± 0.08	+0.32 ± 0.08	6%	1.2%
unix MiniLM	root	human answer (ctrl)	>0.95	[0.64, 0.95]	+0.13 ± 0.08	+0.02 ± 0.13	4%	1.5%
unix MiniLM	root	question + human answer (ctrl)	>0.95	[0.95, 0.95]	+0.41 ± 0.12	+0.16 ± 0.14	9%	3.1%
unix MiniLM	content	answer	0.78	[0.56, 0.95]	+0.27 ± 0.08	-0.08 ± 0.13	8%	3.2%
unix MiniLM	content	question + answer	>0.95	[0.92, 0.95]	+0.35 ± 0.10	+0.09 ± 0.14	10%	3.2%
unix MiniLM	content	answer + thread	>0.95	[0.95, 0.95]	+0.37 ± 0.08	+0.28 ± 0.08	6%	1.2%
unix MiniLM	content	human answer (ctrl)	0.81	[0.52, 0.95]	+0.13 ± 0.08	-0.05 ± 0.13	4%	1.5%
unix MiniLM	content	question + human answer (ctrl)	>0.95	[0.86, 0.95]	+0.37 ± 0.13	+0.11 ± 0.13	9%	3.1%
unix BM25	root	question only (pointer)	>0.95	[0.55, 0.95]	+0.06 ± 0.04	+0.04 ± 0.05	2%	0.4%
unix BM25	root	question + thread (pointer)	>0.95	[0.95, 0.95]	+0.12 ± 0.05	+0.09 ± 0.05	9%	0.4%
unix BM25	root	answer	0.40	[0.21, 0.95]	+0.07 ± 0.07	-0.03 ± 0.07	5%	2.4%
unix BM25	root	question + answer	>0.95	[0.39, 0.95]	+0.14 ± 0.09	+0.01 ± 0.07	7%	2.8%
unix BM25	root	answer + thread	>0.95	[0.95, 0.95]	+0.17 ± 0.06	+0.13 ± 0.04	7%	1.3%
unix BM25	root	human answer (ctrl)	<0.10	[0.10, 0.30]	-0.01 ± 0.01	-0.12 ± 0.04	2%	1.3%
unix BM25	root	question + human answer (ctrl)	0.68	[0.37, 0.94]	+0.08 ± 0.05	-0.04 ± 0.04	4%	1.4%
unix BM25	content	answer	0.35	[0.13, 0.93]	+0.05 ± 0.06	-0.05 ± 0.07	5%	2.4%
unix BM25	content	question + answer	0.89	[0.35, 0.95]	+0.11 ± 0.08	-0.01 ± 0.06	7%	2.8%
unix BM25	content	answer + thread	>0.95	[0.95, 0.95]	+0.14 ± 0.05	+0.11 ± 0.04	7%	1.3%
unix BM25	content	human answer (ctrl)	<0.10	[0.10, 0.30]	-0.01 ± 0.01	-0.12 ± 0.04	2%	1.3%
unix BM25	content	question + human answer (ctrl)	0.68	[0.37, 0.93]	+0.08 ± 0.05	-0.04 ± 0.04	4%	1.4%
webmasters bge	root	question only (pointer)	0.14	[0.12, 0.19]	+0.19 ± 0.16	-1.79 ± 0.28	72%	31.3%
webmasters bge	root	question + thread (pointer)	0.32	[0.14, 0.66]	+0.07 ± 0.07	-0.12 ± 0.09	24%	3.8%
webmasters bge	root	answer	0.36	[0.24, 0.46]	+0.19 ± 0.11	-0.36 ± 0.14	31%	8.1%
webmasters bge	root	question + answer	0.58	[0.36, 0.77]	+0.18 ± 0.07	-0.15 ± 0.09	18%	5.0%
webmasters bge	root	answer + thread	0.40	[0.27, 0.53]	+0.14 ± 0.09	-0.32 ± 0.11	29%	9.1%
webmasters bge	root	human answer (ctrl)	>0.95	[0.38, 0.95]	+0.32 ± 0.20	+0.06 ± 0.20	22%	4.8%
webmasters bge	root	question + human answer (ctrl)	0.30	[0.10, 0.84]	+0.07 ± 0.19	-0.19 ± 0.10	19%	4.4%
webmasters bge	content	answer	0.34	[0.20, 0.45]	+0.16 ± 0.11	-0.38 ± 0.12	31%	8.1%
webmasters bge	content	question + answer	0.46	[0.30, 0.70]	+0.16 ± 0.07	-0.19 ± 0.08	18%	5.0%
webmasters bge	content	answer + thread	0.34	[0.25, 0.46]	+0.11 ± 0.09	-0.39 ± 0.11	29%	9.1%
webmasters bge	content	human answer (ctrl)	0.79	[0.36, 0.95]	+0.31 ± 0.19	-0.05 ± 0.16	22%	4.8%
webmasters bge	content	question + human answer (ctrl)	0.29	[0.10, 0.82]	+0.06 ± 0.19	-0.22 ± 0.09	19%	4.4%
webmasters MiniLM	root	question only (pointer)	0.77	[0.56, 0.92]	+0.98 ± 0.32	-0.32 ± 0.26	42%	14.2%
webmasters MiniLM	root	question + thread (pointer)	0.90	[0.26, 0.95]	+0.22 ± 0.09	-0.04 ± 0.14	28%	5.8%
webmasters MiniLM	root	answer	>0.95	[0.95, 0.95]	+0.29 ± 0.14	+0.25 ± 0.14	11%	1.9%
webmasters MiniLM	root	question + answer	>0.95	[0.95, 0.95]	+0.34 ± 0.15	+0.16 ± 0.19	17%	3.4%
webmasters MiniLM	root	answer + thread	>0.95	[0.95, 0.95]	+0.24 ± 0.09	+0.22 ± 0.11	10%	1.0%
webmasters MiniLM	root	human answer (ctrl)	>0.95	[0.95, 0.95]	+0.18 ± 0.07	+0.13 ± 0.09	7%	1.3%
webmasters MiniLM	root	question + human answer (ctrl)	>0.95	[0.70, 0.95]	+0.11 ± 0.07	+0.09 ± 0.12	18%	2.0%
webmasters MiniLM	content	answer	>0.95	[0.95, 0.95]	+0.29 ± 0.14	+0.24 ± 0.13	11%	1.9%
webmasters MiniLM	content	question + answer	>0.95	[0.95, 0.95]	+0.32 ± 0.15	+0.19 ± 0.17	17%	3.4%
webmasters MiniLM	content	answer + thread	>0.95	[0.95, 0.95]	+0.22 ± 0.09	+0.22 ± 0.09	10%	1.0%
webmasters MiniLM	content	human answer (ctrl)	>0.95	[0.95, 0.95]	+0.18 ± 0.07	+0.12 ± 0.10	7%	1.3%
webmasters MiniLM	content	question + human answer (ctrl)	>0.95	[0.93, 0.95]	+0.11 ± 0.07	+0.09 ± 0.12	18%	2.0%
webmasters BM25	root	question only (pointer)	<0.10	[0.10, 0.17]	-0.01 ± 0.03	-0.08 ± 0.03	7%	0.9%
webmasters BM25	root	question + thread (pointer)	<0.10	[0.10, 0.10]	-0.08 ± 0.06	-0.27 ± 0.07	17%	4.5%
webmasters BM25	root	answer	>0.95	[0.95, 0.95]	+0.09 ± 0.03	+0.11 ± 0.06	10%	1.5%
webmasters BM25	root	question + answer	>0.95	[0.95, 0.95]	+0.14 ± 0.05	+0.16 ± 0.07	14%	1.8%
webmasters BM25	root	answer + thread	>0.95	[0.46, 0.95]	+0.09 ± 0.02	+0.09 ± 0.09	12%	1.7%
webmasters BM25	root	human answer (ctrl)	0.16	[0.10, 0.40]	+0.03 ± 0.04	-0.04 ± 0.06	5%	1.5%
webmasters BM25	root	question + human answer (ctrl)	>0.95	[0.17, 0.95]	+0.09 ± 0.07	+0.07 ± 0.09	9%	1.5%
webmasters BM25	content	answer	>0.95	[0.95, 0.95]	+0.07 ± 0.03	+0.11 ± 0.05	10%	1.5%
webmasters BM25	content	question + answer	>0.95	[0.95, 0.95]	+0.13 ± 0.06	+0.16 ± 0.08	14%	1.8%
webmasters BM25	content	answer + thread	0.80	[0.40, 0.95]	+0.08 ± 0.03	+0.01 ± 0.09	12%	1.7%
webmasters BM25	content	human answer (ctrl)	0.12	[0.10, 0.20]	+0.01 ± 0.03	-0.04 ± 0.05	5%	1.5%
webmasters BM25	content	question + human answer (ctrl)	0.47	[0.12, 0.95]	+0.05 ± 0.05	+0.01 ± 0.07	9%	1.5%
mathematica bge	root	question only (pointer)	<0.10	[0.10, 0.11]	-0.20 ± 0.30	-2.58 ± 0.30	65%	39.0%
mathematica bge	root	question + thread (pointer)	0.70	[0.36, 0.95]	+0.11 ± 0.03	-0.02 ± 0.06	11%	2.5%
mathematica bge	root	answer	0.29	[0.23, 0.52]	+0.18 ± 0.10	-0.46 ± 0.20	22%	11.0%
mathematica bge	root	question + answer	<0.10	[0.10, 0.30]	-0.00 ± 0.09	-0.14 ± 0.06	10%	2.9%
mathematica bge	root	answer + thread	0.64	[0.27, 0.79]	+0.15 ± 0.07	-0.07 ± 0.12	17%	4.5%
mathematica bge	root	human answer (ctrl)	0.11	[0.10, 0.20]	+0.01 ± 0.12	-0.71 ± 0.17	22%	13.7%
mathematica bge	root	question + human answer (ctrl)	0.22	[0.10, 0.45]	+0.04 ± 0.08	-0.32 ± 0.10	14%	7.5%
mathematica bge	content	answer	0.25	[0.18, 0.44]	+0.13 ± 0.10	-0.54 ± 0.19	22%	11.0%
mathematica bge	content	question + answer	<0.10	[0.10, 0.28]	-0.00 ± 0.09	-0.17 ± 0.06	10%	2.9%
mathematica bge	content	answer + thread	0.60	[0.27, 0.76]	+0.14 ± 0.07	-0.10 ± 0.12	17%	4.5%
mathematica bge	content	human answer (ctrl)	0.11	[0.10, 0.18]	+0.01 ± 0.12	-0.74 ± 0.16	22%	13.7%
mathematica bge	content	question + human answer (ctrl)	0.22	[0.10, 0.45]	+0.04 ± 0.08	-0.32 ± 0.10	14%	7.5%
mathematica MiniLM	root	question only (pointer)	0.23	[0.16, 0.31]	+0.33 ± 0.22	-1.15 ± 0.27	46%	22.0%
mathematica MiniLM	root	question + thread (pointer)	<0.10	[0.10, 0.15]	-0.01 ± 0.06	-0.49 ± 0.11	17%	7.2%
mathematica MiniLM	root	answer	>0.95	[0.26, 0.95]	+0.12 ± 0.08	+0.01 ± 0.19	12%	4.4%
mathematica MiniLM	root	question + answer	0.70	[0.28, 0.88]	+0.33 ± 0.11	-0.14 ± 0.21	13%	6.0%
mathematica MiniLM	root	answer + thread	>0.95	[0.95, 0.95]	+0.18 ± 0.08	+0.13 ± 0.11	9%	1.7%
mathematica MiniLM	root	human answer (ctrl)	0.20	[0.10, 0.42]	+0.03 ± 0.06	-0.18 ± 0.07	8%	3.2%
mathematica MiniLM	root	question + human answer (ctrl)	0.68	[0.27, 0.93]	+0.20 ± 0.09	-0.10 ± 0.15	12%	4.7%
mathematica MiniLM	content	answer	0.90	[0.25, 0.95]	+0.11 ± 0.08	-0.01 ± 0.20	12%	4.4%
mathematica MiniLM	content	question + answer	0.65	[0.28, 0.84]	+0.32 ± 0.12	-0.15 ± 0.21	13%	6.0%
mathematica MiniLM	content	answer + thread	>0.95	[0.94, 0.95]	+0.16 ± 0.06	+0.07 ± 0.10	9%	1.7%
mathematica MiniLM	content	human answer (ctrl)	0.20	[0.10, 0.44]	+0.04 ± 0.06	-0.15 ± 0.06	8%	3.2%
mathematica MiniLM	content	question + human answer (ctrl)	0.68	[0.25, 0.94]	+0.20 ± 0.09	-0.10 ± 0.15	12%	4.7%
mathematica BM25	root	question only (pointer)	>0.95	[0.95, 0.95]	+0.00 ± 0.00	+0.00 ± 0.00	3%	0.0%
mathematica BM25	root	question + thread (pointer)	<0.10	[0.10, 0.30]	-0.01 ± 0.03	-0.04 ± 0.04	11%	0.8%
mathematica BM25	root	answer	0.50	[0.10, 0.84]	+0.05 ± 0.07	-0.07 ± 0.11	7%	4.5%
mathematica BM25	root	question + answer	0.52	[0.12, 0.87]	+0.08 ± 0.09	-0.06 ± 0.09	10%	5.0%
mathematica BM25	root	answer + thread	0.50	[0.10, 0.93]	+0.04 ± 0.10	-0.08 ± 0.10	9%	4.2%
mathematica BM25	root	human answer (ctrl)	<0.10	[0.10, 0.10]	-0.06 ± 0.05	-0.04 ± 0.07	3%	2.0%
mathematica BM25	root	question + human answer (ctrl)	>0.95	[0.10, 0.95]	+0.03 ± 0.05	+0.06 ± 0.05	5%	1.4%
mathematica BM25	content	answer	0.35	[0.10, 0.54]	+0.02 ± 0.06	-0.09 ± 0.10	7%	4.5%
mathematica BM25	content	question + answer	0.40	[0.10, 0.60]	+0.05 ± 0.08	-0.07 ± 0.09	10%	5.0%
mathematica BM25	content	answer + thread	0.52	[0.10, 0.90]	+0.04 ± 0.09	-0.06 ± 0.10	9%	4.2%
mathematica BM25	content	human answer (ctrl)	<0.10	[0.10, 0.10]	-0.06 ± 0.05	-0.08 ± 0.05	3%	2.0%
mathematica BM25	content	question + human answer (ctrl)	>0.95	[0.10, 0.95]	+0.03 ± 0.05	+0.01 ± 0.05	5%	1.4%

Table 9: Countermeasures by configuration, content-aware user: Δ (points) at $p_w = 0.1$ and 0.95 and the crossover.

config		*naive write-back*	*reopen, oracle (Guthikonda, 2026)*	*reopen after 1 round*	*reopen after 2 rounds*	*reopen after 1 round, 10% false*	*reopen after 1 round, 20% false*	*depth cap (depth 1 only)*	*regenerate from root*	*uniform penalty, $\lambda = 0.5$*	*uniform penalty, $\lambda = 1$*	*depth penalty, $\lambda = 0.5$*	*depth penalty, $\lambda = 1$*
android bge	Δ 0.1	+0.50	+0.42	+0.42	+0.43	+0.44	+0.41	+0.45	+0.53	+0.12	+0.04	+0.11	+0.04
	Δ 0.95	+0.15	+0.29	+0.28	+0.26	+0.09	+0.06	+0.14	+0.20	+0.08	+0.04	+0.06	+0.04
	p^*	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95
android MiniLM	Δ 0.1	+0.57	+0.61	+0.62	+0.60	+0.51	+0.51	+0.55	+0.55	+0.09	+0.02	+0.07	+0.02
	Δ 0.95	+0.14	+0.37	+0.28	+0.22	+0.25	+0.19	+0.18	+0.22	-0.06	-0.04	-0.08	-0.03
	p^*	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	0.29	0.30	0.29	0.35
android BM25	Δ 0.1	+0.12	+0.12	+0.12	+0.12	+0.11	+0.11	+0.12	+0.12	+0.04	+0.00	+0.04	+0.00
	Δ 0.95	+0.18	+0.16	+0.17	+0.17	+0.11	+0.11	+0.17	+0.17	+0.04	+0.00	+0.04	+0.00
	p^*	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95
unix bge	Δ 0.1	+0.40	+0.38	+0.38	+0.38	+0.41	+0.35	+0.39	+0.41	+0.18	+0.02	+0.18	+0.02
	Δ 0.95	+0.21	+0.27	+0.25	+0.24	+0.24	+0.24	+0.20	+0.25	+0.16	+0.02	+0.16	+0.02
	p^*	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95
unix MiniLM	Δ 0.1	+0.35	+0.39	+0.39	+0.38	+0.39	+0.35	+0.33	+0.31	+0.02	+0.01	+0.02	+0.01
	Δ 0.95	+0.09	+0.24	+0.22	+0.16	+0.23	+0.18	+0.08	+0.09	-0.06	+0.01	-0.06	+0.01
	p^*	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	0.24	0.57	0.24	0.57
unix BM25	Δ 0.1	+0.11	+0.13	+0.13	+0.12	+0.09	+0.08	+0.09	+0.08	+0.03	+0.02	+0.04	+0.02
	Δ 0.95	-0.01	+0.06	+0.06	+0.05	+0.05	+0.05	-0.04	-0.09	+0.03	+0.02	+0.04	+0.02
	p^*	0.89	>0.95	>0.95	>0.95	>0.95	>0.95	0.82	0.45	>0.95	>0.95	>0.95	>0.95
webmasters bge	Δ 0.1	+0.16	+0.22	+0.22	+0.22	+0.17	+0.15	+0.16	+0.18	+0.02	+0.00	+0.02	+0.00
	Δ 0.95	-0.19	-0.01	-0.07	-0.12	-0.04	-0.05	-0.18	-0.15	-0.06	+0.00	-0.06	+0.00
	p^*	0.46	0.86	0.81	0.65	0.59	0.58	0.47	0.57	0.22	>0.95	0.22	>0.95
webmasters MiniLM	Δ 0.1	+0.32	+0.31	+0.31	+0.31	+0.26	+0.24	+0.31	+0.31	+0.11	+0.00	+0.11	+0.00
	Δ 0.95	+0.19	+0.27	+0.27	+0.26	+0.22	+0.20	+0.17	+0.16	+0.10	+0.00	+0.10	+0.00
	p^*	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95
webmasters BM25	Δ 0.1	+0.13	+0.13	+0.14	+0.14	+0.10	+0.06	+0.13	+0.13	+0.00	+0.00	+0.00	+0.00
	Δ 0.95	+0.16	+0.13	+0.14	+0.15	+0.14	+0.13	+0.16	+0.15	+0.00	+0.00	+0.00	+0.00
	p^*	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95	>0.95
mathematica bge	Δ 0.1	-0.00	+0.01	+0.01	+0.01	-0.00	+0.01	-0.00	-0.00	+0.00	+0.00	+0.00	+0.00
	Δ 0.95	-0.17	-0.08	-0.10	-0.14	-0.08	-0.08	-0.16	-0.17	+0.04	+0.04	+0.04	+0.04
	p^*	<0.10	0.20	0.23	0.22	<0.10	0.24	<0.10	<0.10	>0.95	>0.95	>0.95	>0.95
mathematica MiniLM	Δ 0.1	+0.32	+0.29	+0.29	+0.27	+0.35	+0.32	+0.32	+0.32	+0.07	+0.04	+0.07	+0.04
	Δ 0.95	-0.15	+0.12	+0.07	+0.00	+0.06	+0.04	-0.13	-0.16	-0.04	+0.04	-0.01	+0.04
	p^*	0.65	>0.95	>0.95	0.69	0.68	0.67	0.65	0.62	0.29	>0.95	0.29	>0.95
mathematica BM25	Δ 0.1	+0.05	+0.07	+0.07	+0.07	+0.10	+0.07	+0.05	+0.05	+0.03	+0.00	+0.03	+0.00
	Δ 0.95	-0.07	+0.03	+0.04	+0.01	-0.00	-0.00	-0.06	-0.07	+0.08	+0.00	+0.08	+0.00
	p^*	0.40	0.57	0.39	0.38	0.67	0.26	0.40	0.40	>0.95	>0.95	>0.95	>0.95